

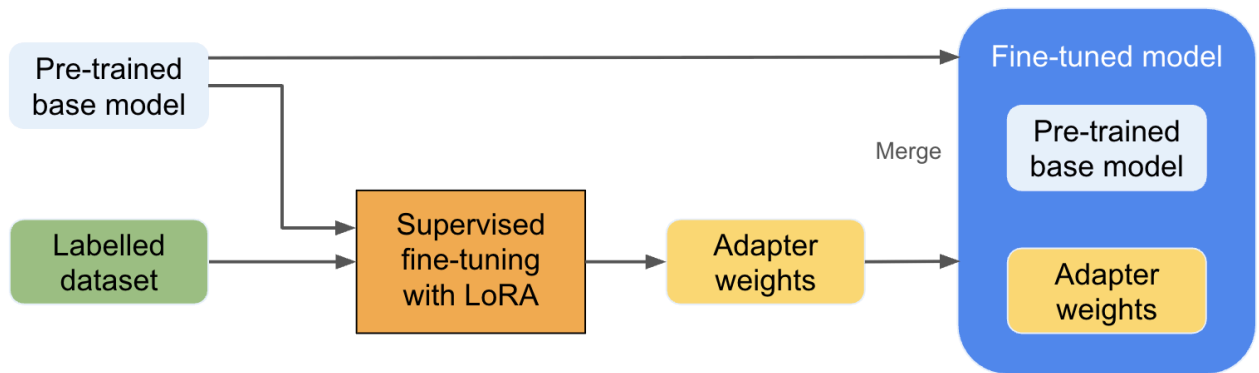


LoRA Adapters

What is LoRA?

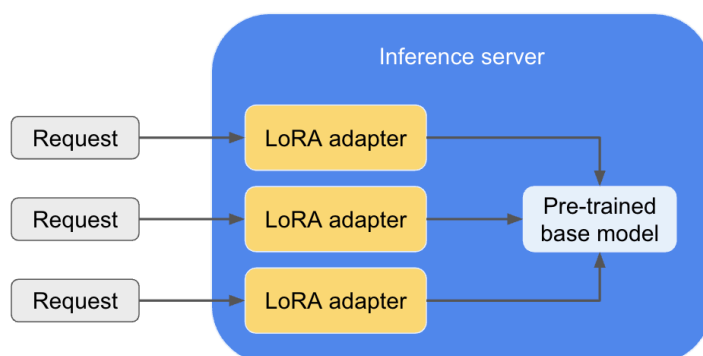
[LoRA \(Low Rank Adaptation\)](#) is a fine-tuning technique that is used to adapt a pre-trained base model to specific tasks without retraining the entire model.

LoRA introduces small adapter layers that are trained on a targeted dataset, while keeping the original base model frozen.



Deploying multiple LoRA Adapters

You can deploy multiple LoRA adapters with a base model to a containerized platform such as Google Kubernetes Engine (GKE) or Cloud Run.

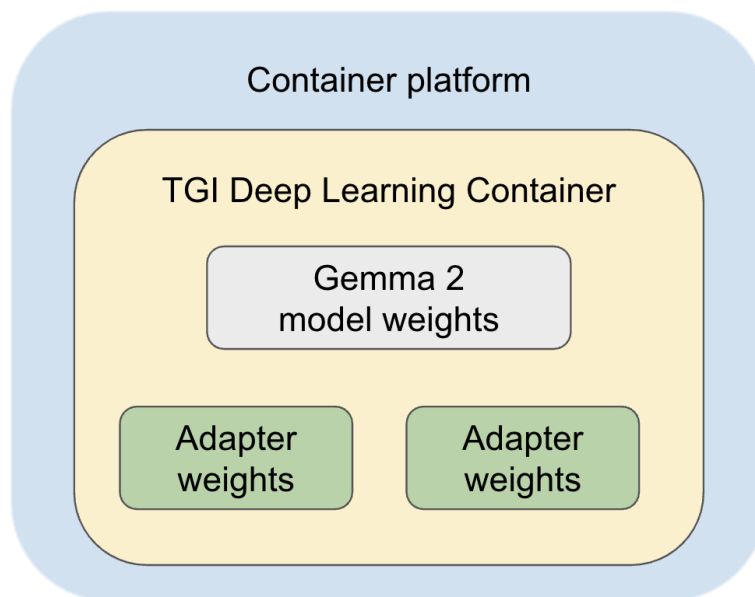


In a multi-LoRA deployment, you deploy a single base model and load multiple LoRA adapters. You can then dynamically select a different adapter for each incoming request. You don't have to create multiple deployments of the model which makes it cost efficient.

Implementing LoRA

To implement LoRA, you can use **TGI** - an open source text generation inference server from [Hugging Face](#).

- You can use the TGI Deep Learning Container from Hugging Face along with Gemma 2, Google's open large language model, and multiple LoRA adapters.
- If using GKE, you'll need to create a Kubernetes cluster where you deploy the container.



- Write your deployment manifest to deploy the TGI container along with the required LoRA adapters, and a load balancer (GKE) to provide access to the container.
- Deploy the container to your platform.

Resources

[YouTube video](#) on deploying Gemma 2 with multiple LoRA adapters on GKE.